

The empirical recurrence for OEIS A297990, recovered from its tail

Adrian Perez Fontelles

Independent researcher

8 September 2026

Abstract

OEIS A297990 records “Empirical recurrence of order 99” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 157$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 6$, so the recurrence is proved for every $n \geq 105$. Its last coefficient is $c_{99} = 512 \neq 0$, so no further tail satisfies anything shorter; any order-99 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A297990 is “Number of nX5 0..1 arrays with every element equal to 1, 2, 3 or 6 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

3, 166, 424, 1446, 5100, 18189, 62390, 213997, ...

The entry states:

Empirical recurrence of order 99 (see link above)

As of the “Last modified” line on the live entry (Jan 10 2018 07:35:06, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 157$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 157$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 99: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 6$ is the first whose minimal order is exactly the stated 99.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 314$ exact terms from $n = 6$ on, it returns order 99 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{99} c_i a(n-i),$$

c_1	6	c_{26}	9164310	c_{51}	204153914	c_{76}	27731792
c_2	−12	c_{27}	−12654516	c_{52}	−183998502	c_{77}	−7891828
c_3	44	c_{28}	19378136	c_{53}	−12646715	c_{78}	−10084486
c_4	−171	c_{29}	−26587808	c_{54}	−297224923	c_{79}	−15766498
c_5	290	c_{30}	28107221	c_{55}	36289303	c_{80}	206027
c_6	−781	c_{31}	−37493873	c_{56}	78441302	c_{81}	331757
c_7	2202	c_{32}	49648946	c_{57}	353574206	c_{82}	5078671
c_8	−3277	c_{33}	−43667475	c_{58}	158813002	c_{83}	−342083
c_9	7657	c_{34}	50464341	c_{59}	6110330	c_{84}	646231
c_{10}	−16207	c_{35}	−60662903	c_{60}	−272068901	c_{85}	−1127354
c_{11}	22764	c_{36}	42659349	c_{61}	−224886937	c_{86}	535307
c_{12}	−46217	c_{37}	−40230976	c_{62}	−93445253	c_{87}	−11160
c_{13}	71718	c_{38}	35046780	c_{63}	114659007	c_{88}	306432
c_{14}	−104780	c_{39}	−4393832	c_{64}	135085920	c_{89}	−192898
c_{15}	173118	c_{40}	226058	c_{65}	61951315	c_{90}	−96442
c_{16}	−172610	c_{41}	41897800	c_{66}	−25279745	c_{91}	−101610
c_{17}	303223	c_{42}	−67966211	c_{67}	−29280081	c_{92}	16668
c_{18}	−337776	c_{43}	58299378	c_{68}	21945670	c_{93}	29312
c_{19}	24118	c_{44}	−149766341	c_{69}	28233479	c_{94}	19808
c_{20}	−334446	c_{45}	131827539	c_{70}	3493280	c_{95}	5012
c_{21}	−186200	c_{46}	−128091043	c_{71}	−53654552	c_{96}	−4136
c_{22}	1554901	c_{47}	210442265	c_{72}	−43749778	c_{97}	−1312
c_{23}	−1478550	c_{48}	−161153036	c_{73}	−22380880	c_{98}	−1216
c_{24}	3541634	c_{49}	193766993	c_{74}	32266024	c_{99}	512
c_{25}	−7071637	c_{50}	−140890223	c_{75}	30994624		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 105$, and it is the recurrence OEIS A297990 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 6$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{99} - c_1x^{99-1} - \dots - c_{99}$. Its constant term is $-c_{99} = -512$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 99 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 99, so $\deg q = 99$, and it is monic. It holds from some index t on. From $\max(t, 6)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 99, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 23 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 99, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 512.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-99 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A297990.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.